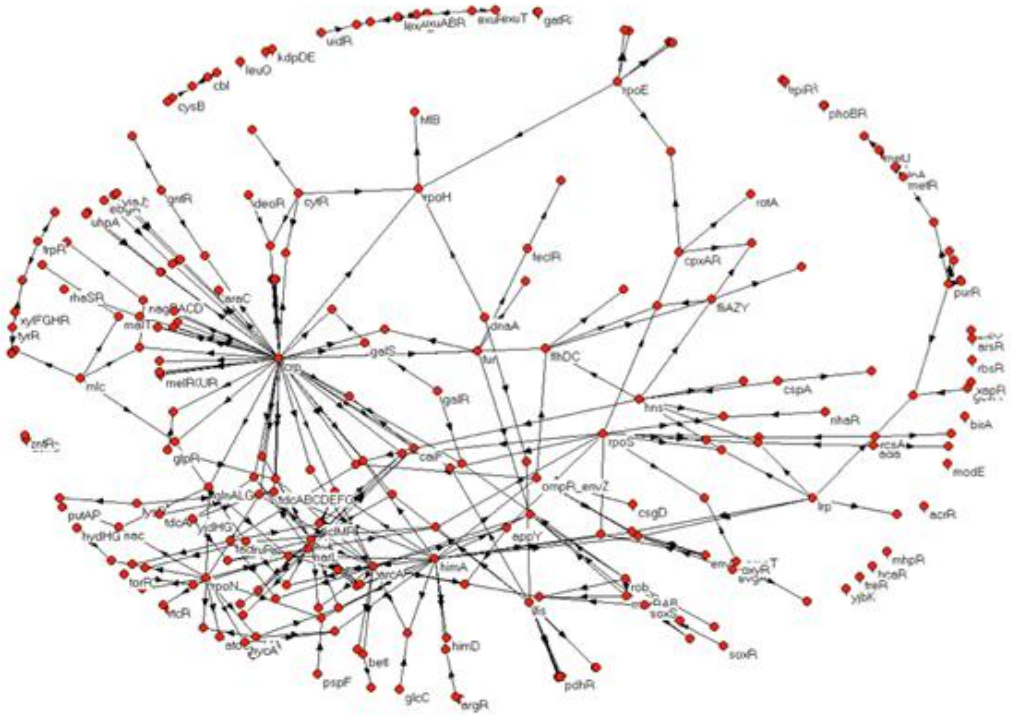
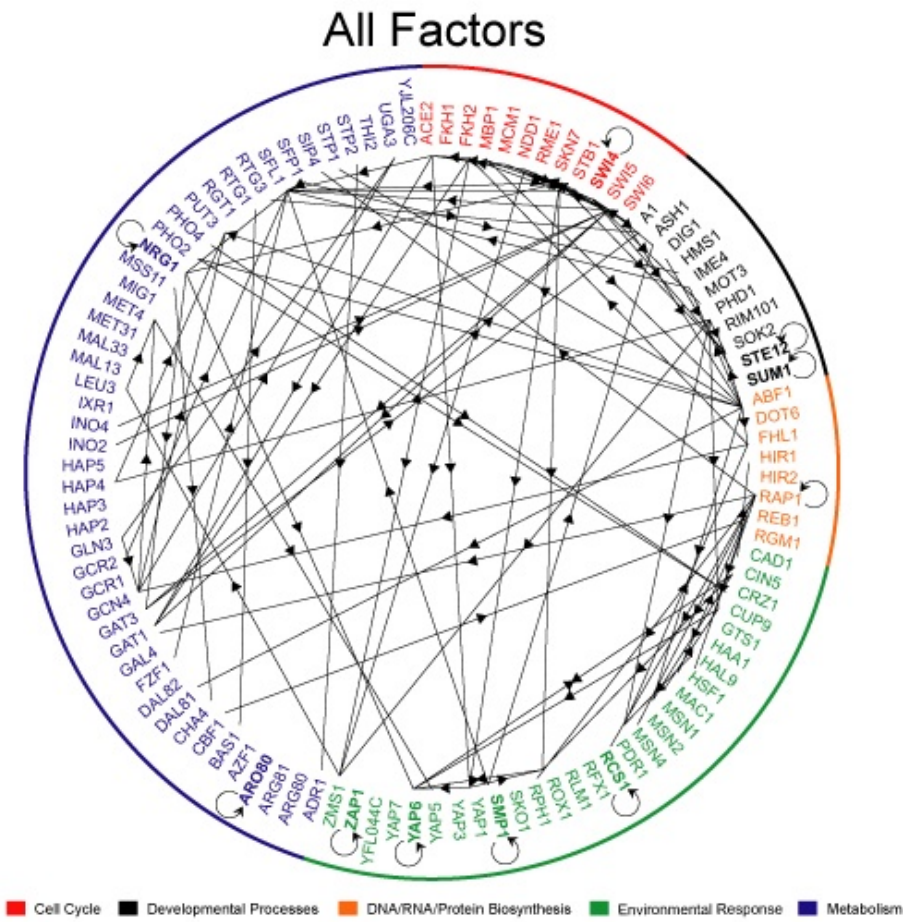
Lecture 17: Gene Regulation Networks



Gene regulation, chemical equations, motifs

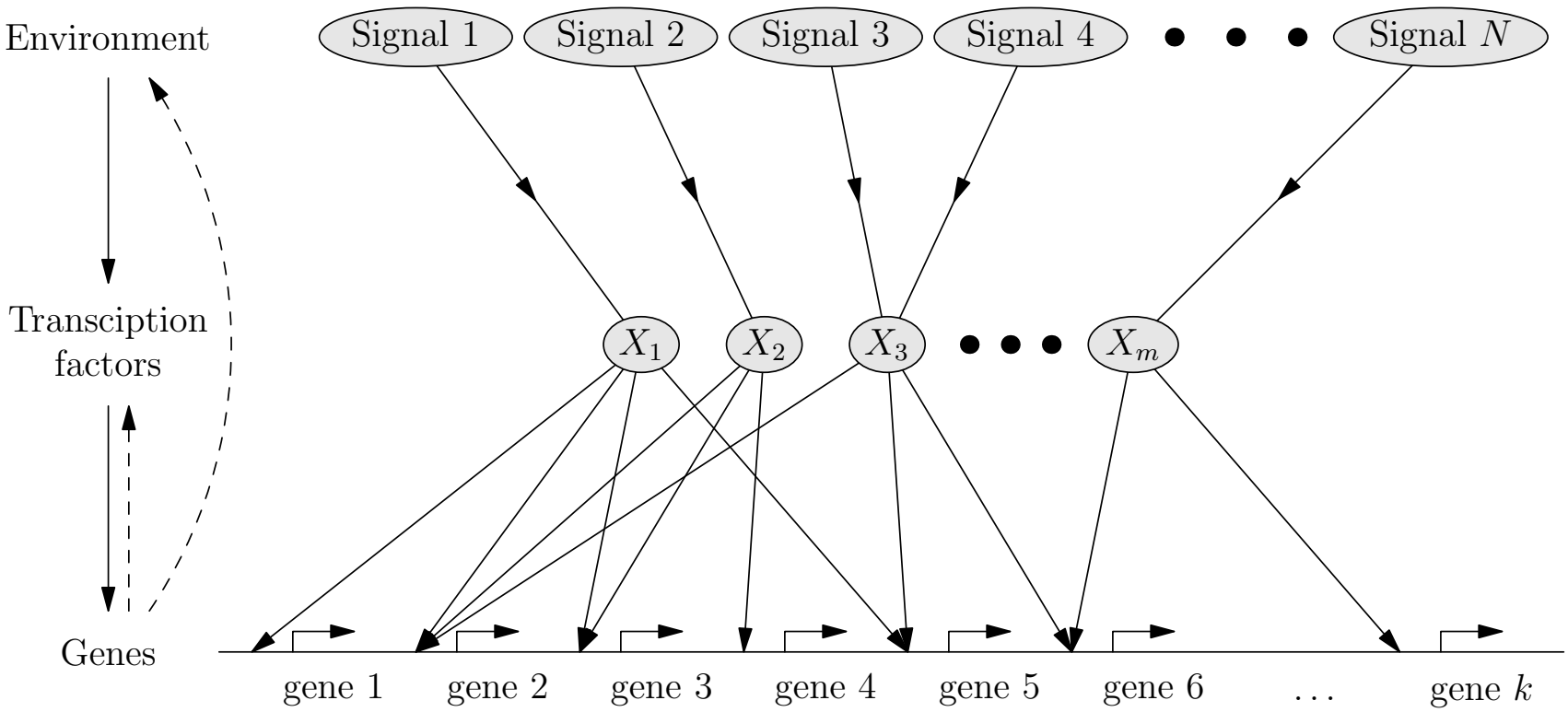
Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs

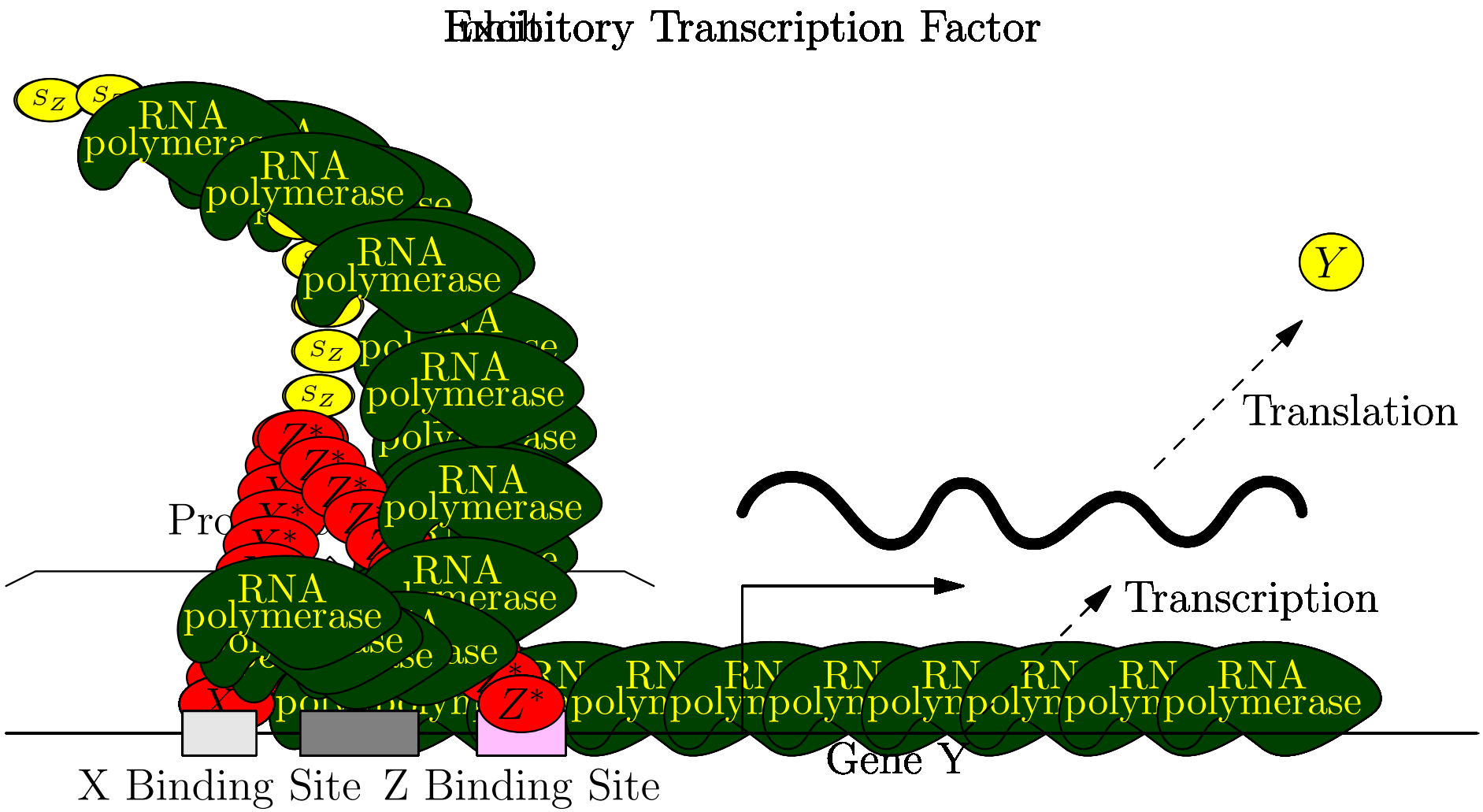


Transciption Networks

- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



Gene Transcription Regulation



Time Scales

- Reactions happen on different time scales
  - ⋆ Binding of small (signalling) molecules: ~ 1 msec
  - ⋆ Binding of active transcription factor: ~ 1 sec
  - ⋆ Transciption + translation of gene: ~ 5 min
  - ⋆ 50% change in concentration of protein: ~ 1 hour
- We can assume all events at earlier time scales happen instantly

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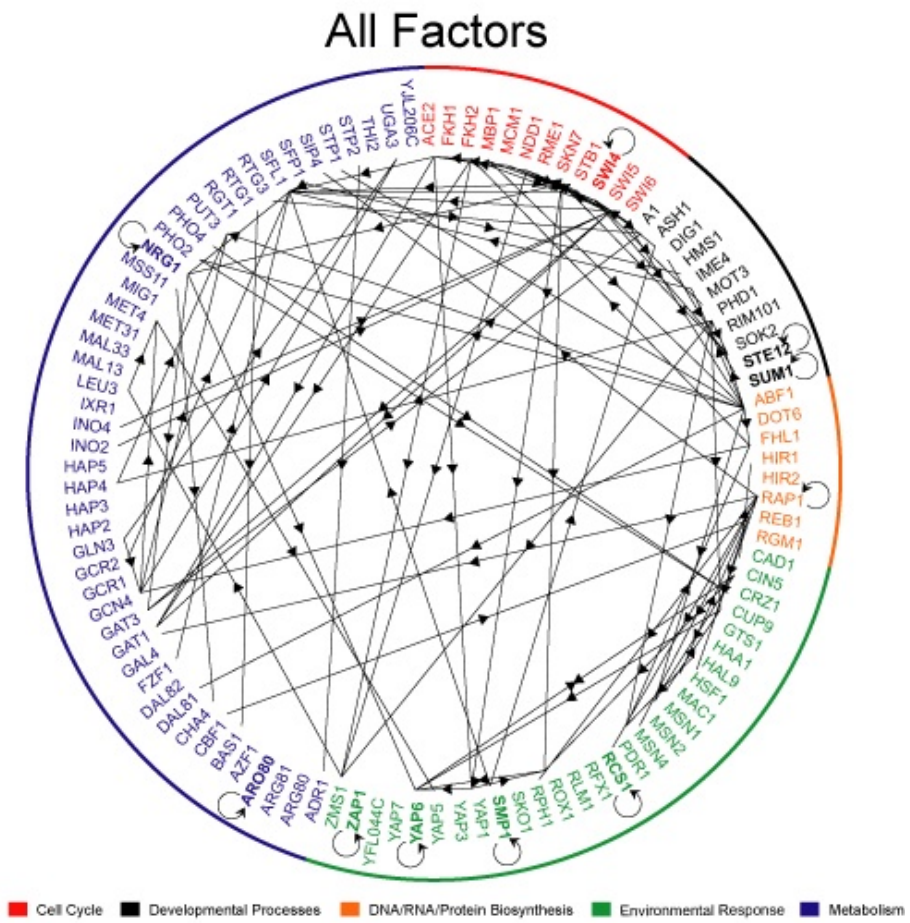


Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations

$$X \xrightarrow{k_1} X + RNA_Y$$

$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$

$$RNA_Y \xrightarrow{k_3} \varnothing$$

- These are massively simplified (but we can ignore reactions that happen quickly)

- The rate of reaction depends on the number/concentration of the molecules on the left-hand side and the reaction rate k_i

Changes on Average

- Chemical equations happen at random, so the rate equations describe the average changes

$$X \xrightarrow{k_1} X + RNA_Y$$

$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$

$$RNA_Y \xrightarrow{k_3} \varnothing$$

- Let $[X]$ denote the (mean) concentration of X then

$$\frac{\mathrm{d}[RNA_Y]}{\mathrm{d}t} = k_1 [X] - k_3 [RNA_Y]$$

$$\frac{\mathrm{d}[Y]}{\mathrm{d}t} = k_2 [RNA_Y]$$

Reaction Rates and Concentrations

- When we consider reaction involving multiple molecular species, e.g. two proteins forming a complex

$$X + Y \xrightarrow{k_4} C$$

- The reaction rates depend on concentrations of X and Y (how likely is it for two molecules to collide)
- To find the concentration we have to divide by the volume of the cell, V
- To obtain concentrations in moles (we also have to divide by Avogadro's number A)

Changes on Average

- Thus

$$X + Y \xrightarrow{k_4} C$$

- Implies on average

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = \frac{\mathrm{d}[Y]}{\mathrm{d}t} = -k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}X}{\mathrm{d}t} = \frac{\mathrm{d}Y}{\mathrm{d}t} = -\frac{k_4}{\Omega}\,X\,Y$$

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}C}{\mathrm{d}t} = \frac{k_4}{\Omega}\,X\,Y$$

- $\Omega = V\,A$, $[X] = X/\Omega$ and $[Y] = Y/\Omega$

Fluctuations

- In chemistry we are used to there being so many molecules that the average rate equations provide a very good approximation of what really happens
- This is often true in biology, but not always
- Cells are small and some proteins can occur in very small numbers
- We return to the problem of modelling systems with small number of molecules in the last lecture
- For the time being we take the differential equation model as an accurate description of our gene regulation network

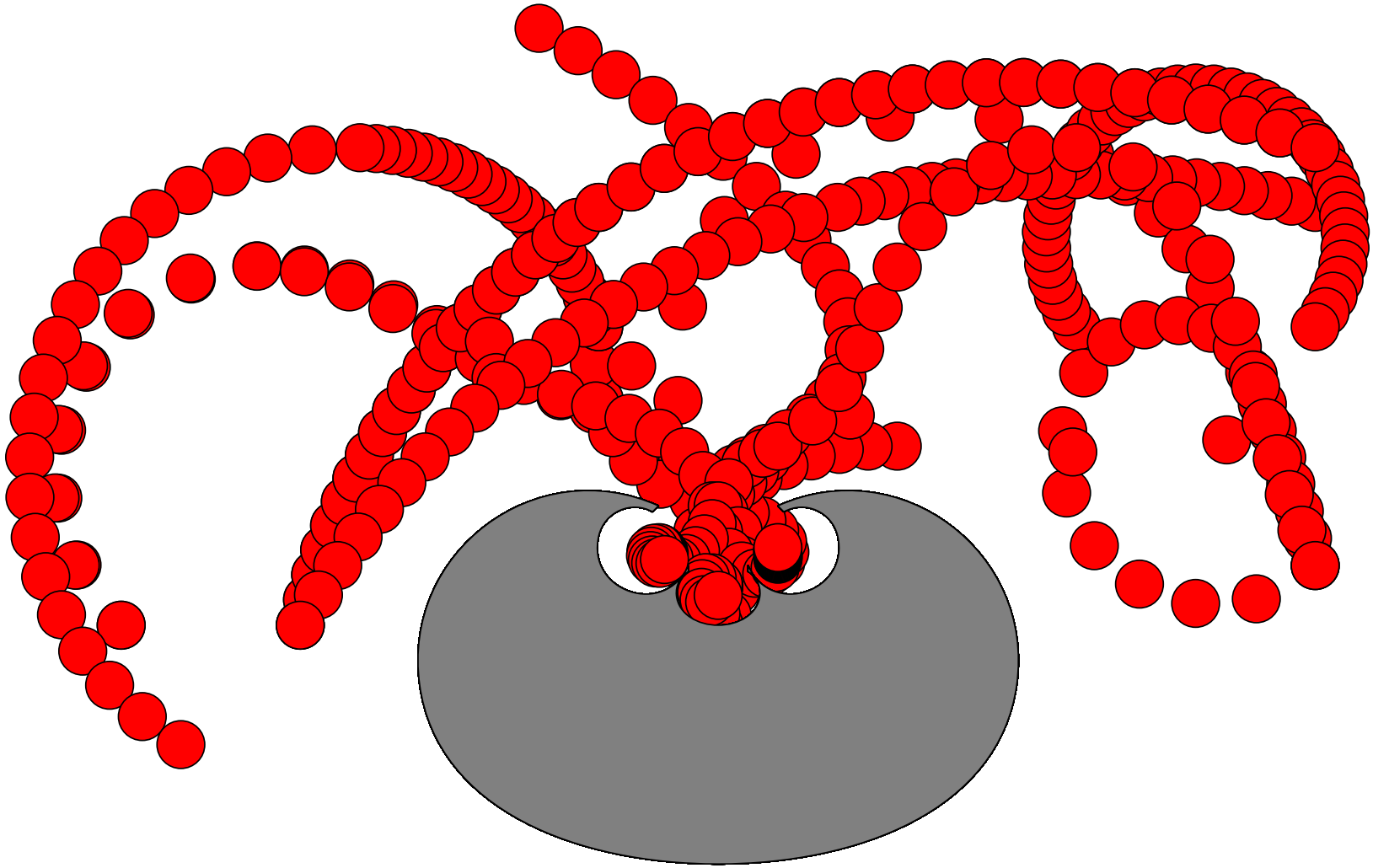
Switches

- Our molecular equation depend on the product of concentrations of the components on the left-hand side
- This can lead to strong non-linearities
- Such non-linearity can simulate switch like mechanism
- This is often useful for robust regulation

Signalling

- We showed earlier that proteins are often *activated* by signalling molecules combining with the protein to form a complex
- Proteins can have multiple sub-units that each bind with a signalling molecule
- A complex might only be stable when the protein has bound some number of signalling molecules

Multi-signalling



Hill Equation

- We assume that we form a complex with n signalling molecules S_X

$$n\,S_X + X \xrightarrow{k_{on}} C \qquad\qquad C \xrightarrow{k_{off}} n\,S_X + X$$

- Which we can model using

$$\frac{d[C]}{dt} = k_{\text{on}}[X][S_X]^n - k_{\text{off}}[C]$$

$$(K_X^n + [S_X]^n)[C] = [X][S_X]^n$$

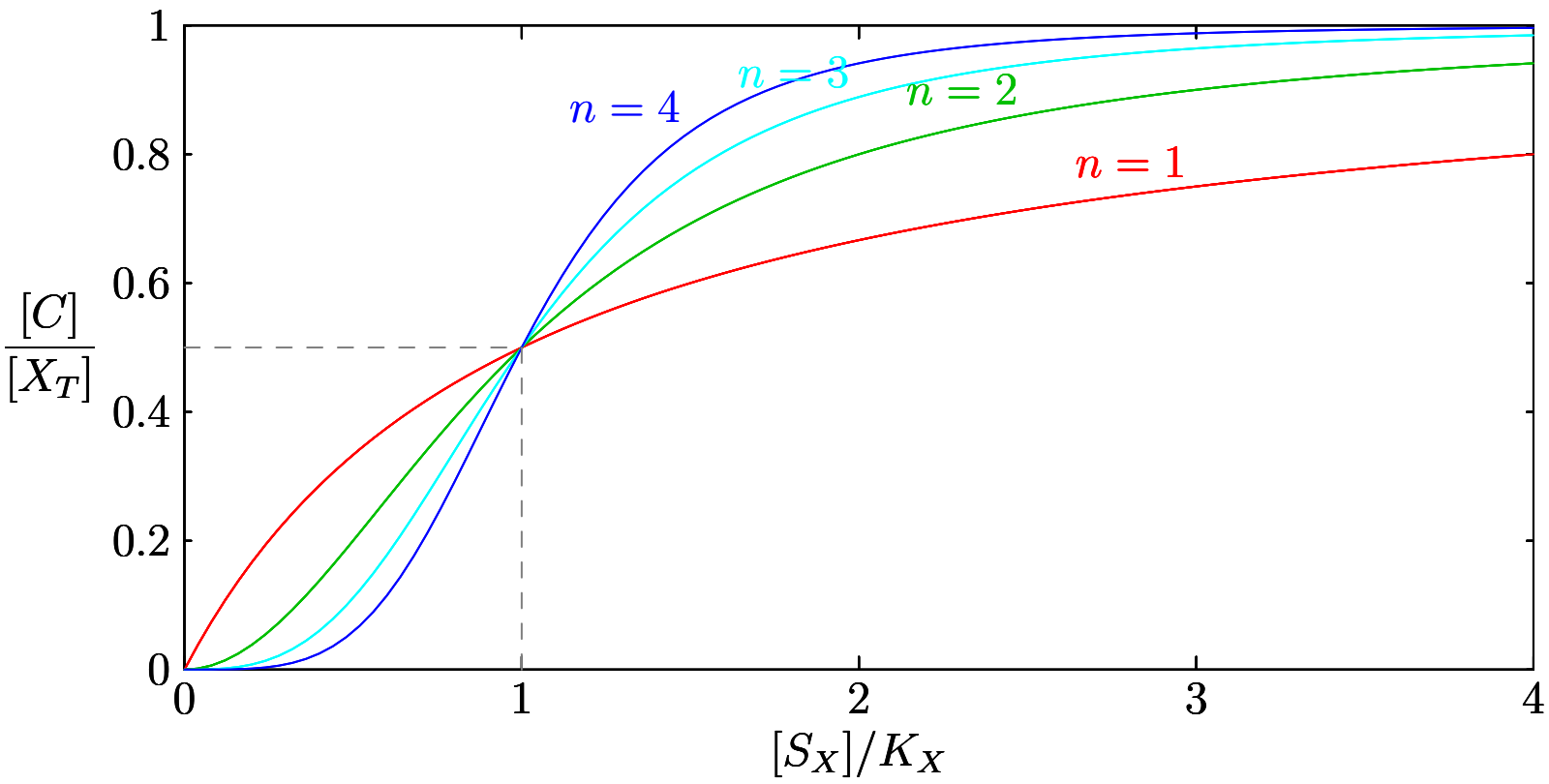
$$[C] = \frac{[X][S_X]^n}{K_X^n + [S_X]^n}$$

$$k_{\text{on}}[X][S_X]^n - k_{\text{off}}[C] = 0$$

But we need to know the Hill coefficients

Hill Coefficient

$$\frac{[C]}{[X_T]} = \frac{[S_X]^n}{K_X^n + [S_X]^n}$$



Activator and Repressor

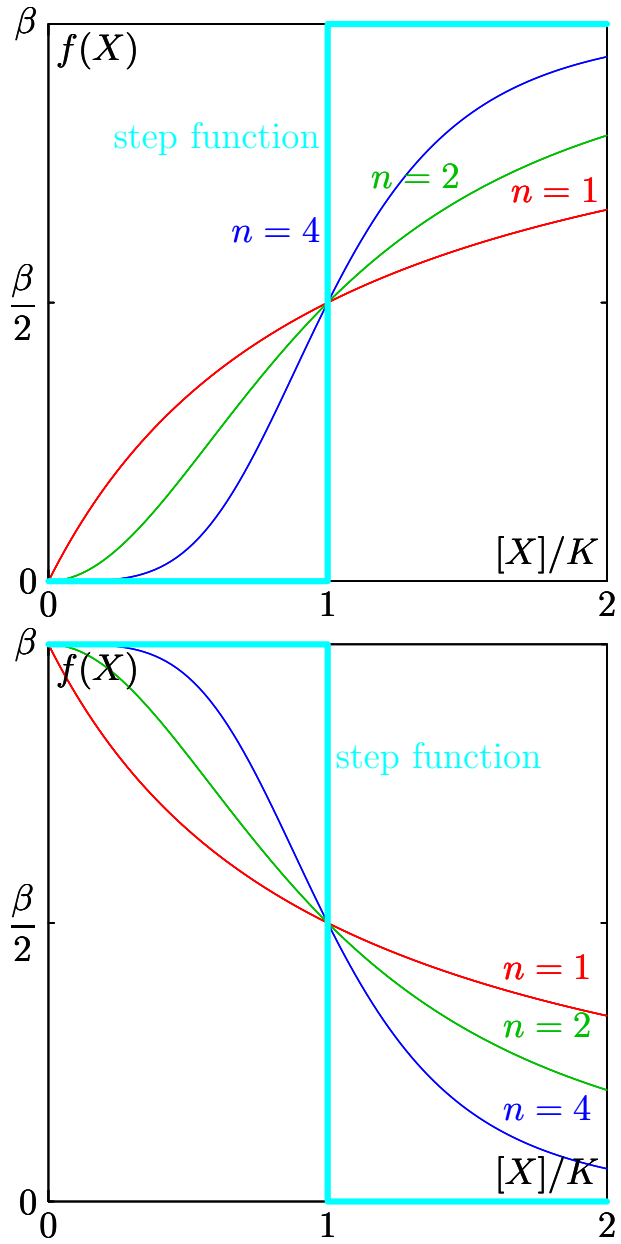
- A protein that acts to increase the expression of a protein we call an **activator**

- The activation coefficient is often given by the Hill-equation

$$f([X]) = \frac{\beta\,[X]^n}{K^n + [X]^n}$$

- By a similar argument a protein that acts as a **repressor** is given by

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n}$$



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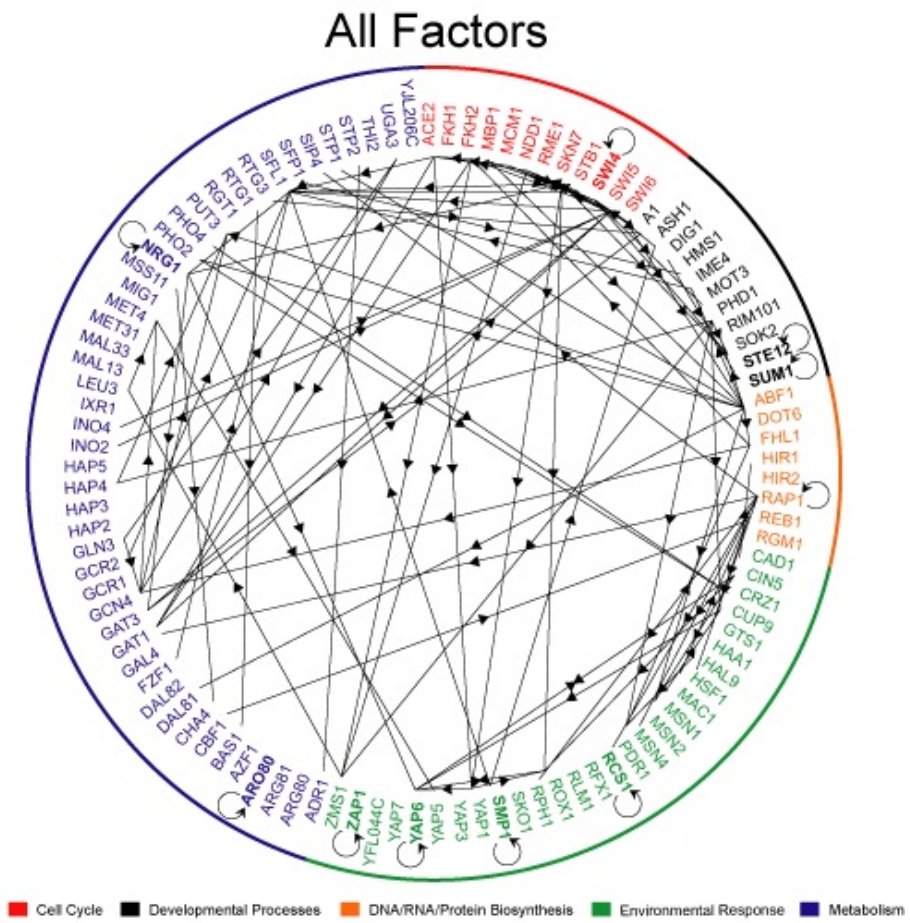
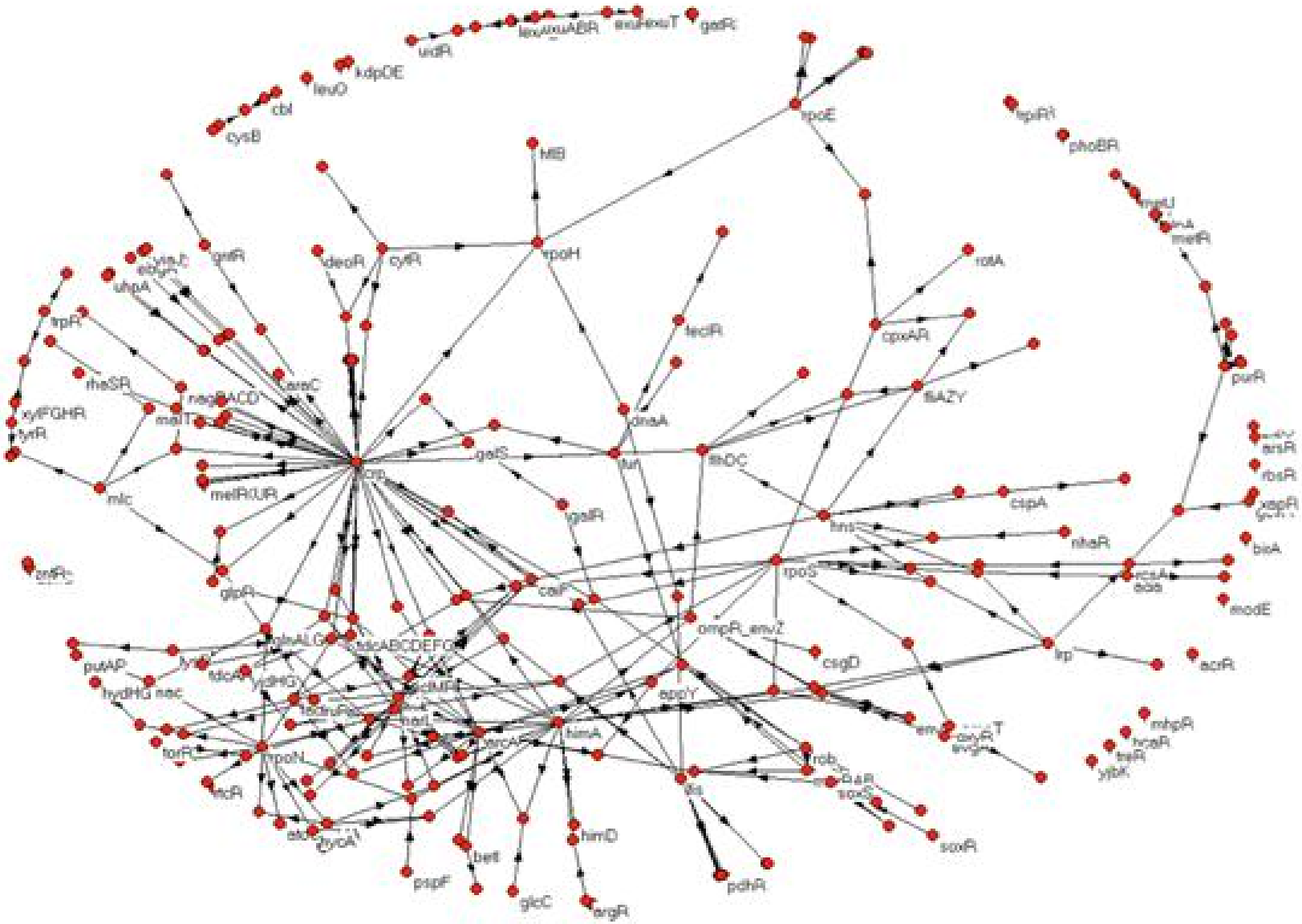


Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture
- Gene regulation networks tend to be rather complicated
- To make sense of them we wish to look for patterns, "**motifs**", that occur very often
- We look at interactions that occur more often than would be expected to happens in a random network

E-Coli GRN


gadR
uidR
kdpDE
leuO
cbl
cysB
rpoE
phoBR
hflB
rpoH
deoR
cytR
rotA
tecR
cpxAR
fliAZY
dnaA
fur
fhuDC
galS
galR
crp
melR(UR)
mlc
rhaSR
xylFGHR
malT
ansR
rbsR
xapR
bioA
modE
ctpA
rhaR
rpoS
cstA
ompR_envZ
csgD
acrR
putAP
hydHG
nac
torR
rpoN
fecR
himA
himD
bet
pspF
glcC
argR
pdhR
soxS
soxR
mhpR
hcaR
ybtK

Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is N^2
- The number of self edges is N so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose E edges k will be self-edges is

$$\mathbb{P}\left(k|E,p\right) = \mathrm{Binom}(k|E,p) = \binom{E}{k} p^k \left(1-p\right)^{E-k}$$

- This has mean $\mathbb{E}\left[N_{\text{self}}\right] = E\,p \approx 1.2$ and standard deviation $\sigma = \sqrt{E\,p\left(1-p\right)} \approx 1.1$
- In fact we have $N_{\text{self}} = 40$ (that is, $32\,\sigma$ above the mean)

No Auto-Regulation

- To understand why auto-regulation is interesting let us first consider the case with no negative feedback

- We assume

$$\varnothing \xrightarrow{\hat{\beta}} \hat{X}$$

$$\hat{X} \xrightarrow{\hat{\alpha}} \varnothing$$

- So on average

$$\frac{\mathrm{d}[\hat{X}]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}]$$

- This has a steady state $[\hat{X}_{eq}] = \hat{\beta}/\hat{\alpha}$

Solving Differential Equations

- The full dynamics is given by the solution to the differential equation and depend on the initial condition, $[\hat{X}(0)]$
- Most differential equations cannot be solved in closed form
- However, they are usually quite easy to solve numerically (though it is difficult to understand how the solution depend on the parameters)
- Our equation is linear and is easy to solve. For $[\hat{X}(0)] = 0$

$$\frac{\mathrm{d}[\hat{X}(t)]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}(t)] \qquad \Rightarrow \qquad [\hat{X}(t)] = \frac{\hat{\beta}}{\hat{\alpha}}\left(1 - \mathrm{e}^{-\hat{\alpha}\,t}\right)$$

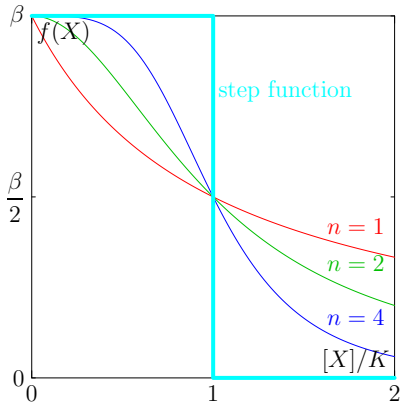
Auto-regulation

- We assume a negative feedback so that

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = f([X]) - \alpha\,[X]$$

- Where we have a Hill-type repressor

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n} \approx \beta\ [\![[X] < K]\!]$$



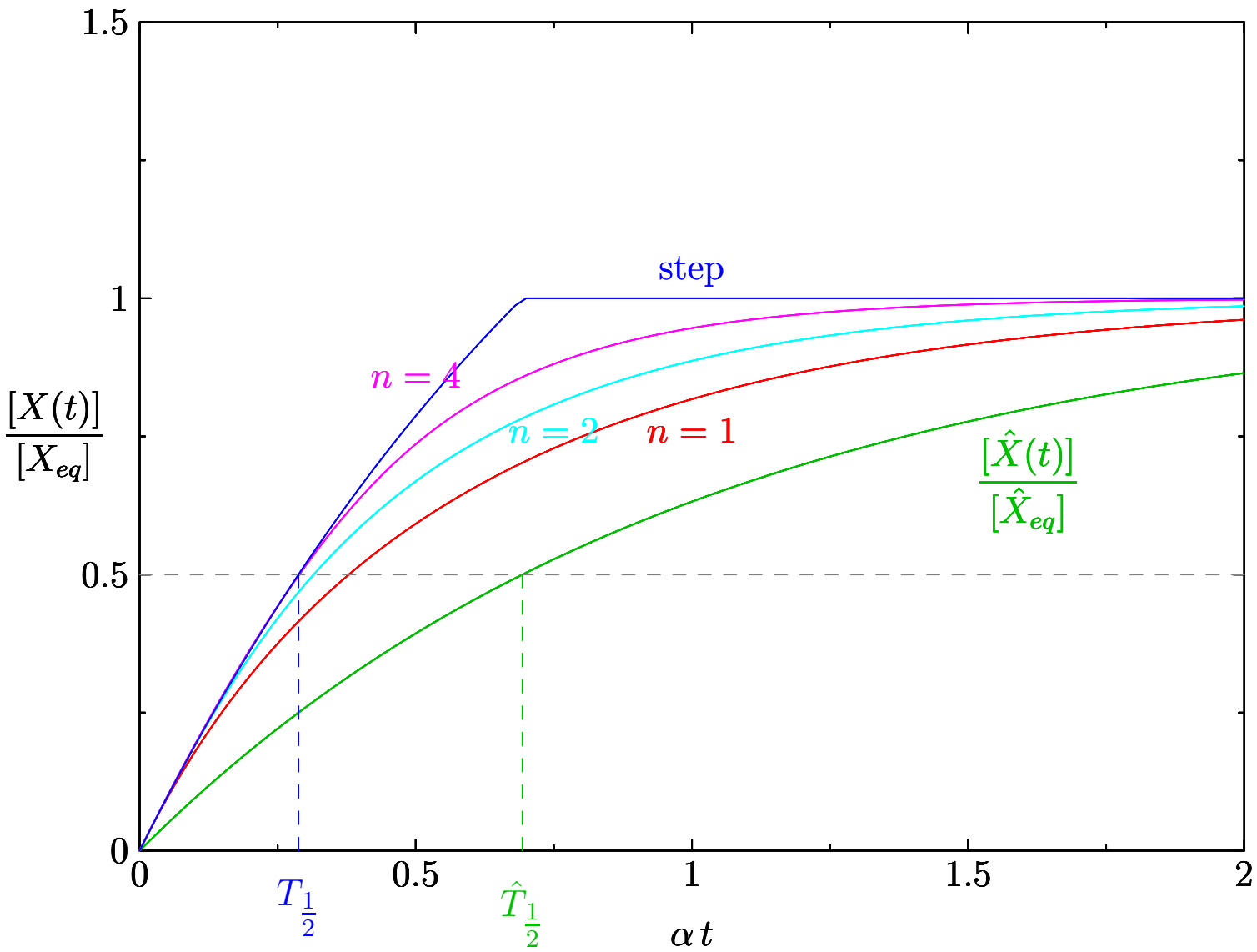
- Starting from $[X(0)] = 0$ initially $[X(t)] = \beta\, t$

- When $[X(t)] > K$ then $f([X(t)]) = f(K + \epsilon) \approx 0$ and a steady state is reached

Controlled Comparison

- To compare to having no feedback we want our two circuits to be as similar as possible
- Choose the same decay rate $\alpha = \hat{\alpha}$
- Choose $K = \hat{\beta}/\hat{\alpha}$ so that the steady state or equilibrium value of $X(t)$ is the same in both systems $[\hat{X}_{eq}] = [X_{eq}]$
- Note that in the auto-regulation system the rate of production β is decoupled from the steady-state

$\beta = 2$


1.5
1
0.5
0
0
0.5
1
1.5
2
step
$n = 4$
$n = 2$
$n = 1$
$\frac{[X(t)]}{[X_{eq}]}$
$\frac{[\hat{X}(t)]}{[\hat{X}_{eq}]}$
$T_{\frac{1}{2}}$
$\hat{T}_{\frac{1}{2}}$
αt

Advantages of Auto-regulation

- Auto-regulation allows us to use a large production rate β for the same equilibrium, speeding up the equilibration time
- Note that we can independent evolve K (by mutations to the binding site of X in the promoter region) and β, by mutations in the binding site of RNA polymerase
- Another advantage is that β and α can strongly fluctuate during the life time of a cell caused by fluctuations in metabolic capacity
- K depends on the chemical bonds in the binding site and is typically more stable

Summary

- We can begin to understand some of the gene regulation dynamics by writing down chemical equations
- We can express the mean change in number of molecules through differential equations
- We can find common motifs, by looking for patterns that occur in GRN more often than we would expect in a random network
- One common motif is negative auto-regulation (negative feedback) that allows us to increase the response rate to reach the same equilibrium level of a molecule